Report: **AM2024-0074**

TEM investigation on particle samples of natural origin

Short Summary 1

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Documentation Overview

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Order of the Client	PO 4500228008 dated 26 July 2024

Place of Performance and Processing Time

Place of Performance	Center for Applied Analytics Fraunhofer Institute for Silicate Research Neunerplatz 2 97082 Würzburg Germany	
Processing Time	Sample receipt	29.07.2024
	Begin of the analysis	31.07.2024
	End of the analysis	ongoing
	Date of report	06.08.2024

Sample Overview

Sample No.	Description
P01	Graphene – Batch: MB2955A2 – Date: 24.07.2024 (de)

P01

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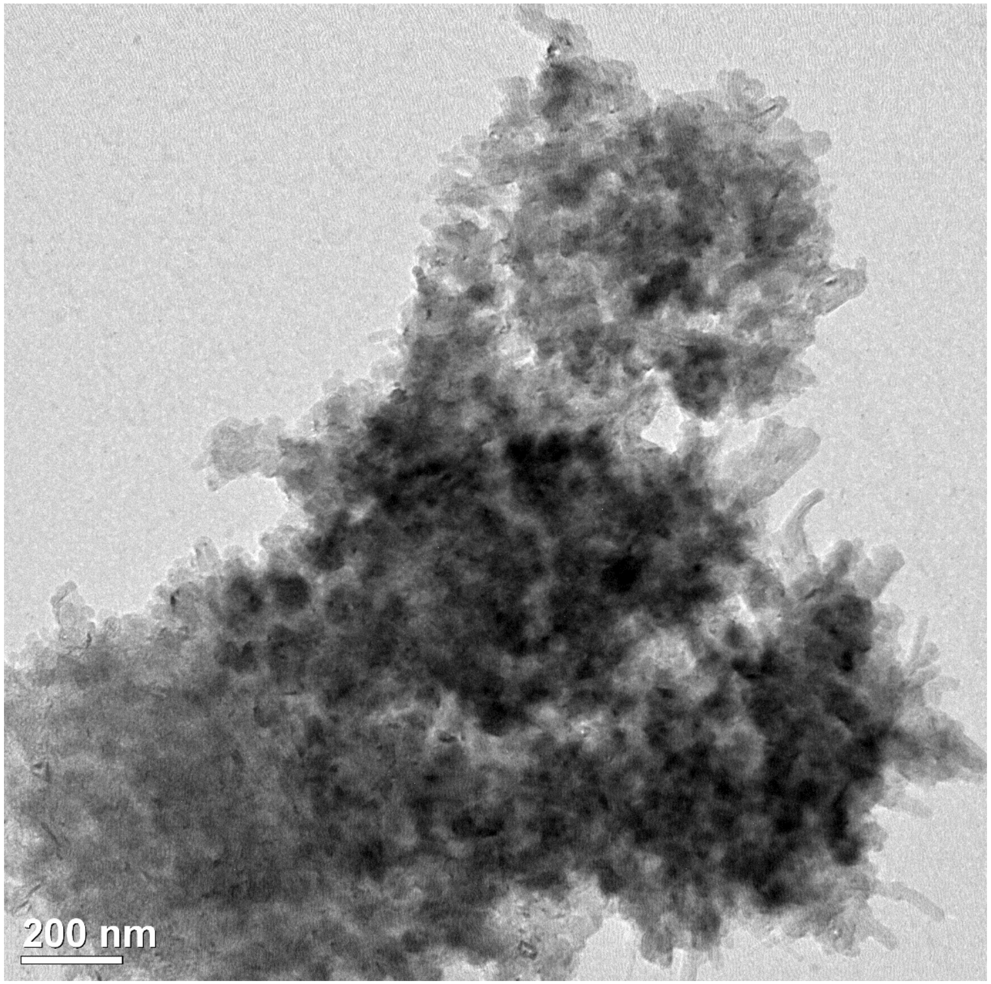
Procedure of Testing

- The received sample was ground before a small amount of the powder was dispersed in 1 part IPA and 4 parts.
- The dispersion was treated in an ultrasonic bath for 20 min.
- The dispersion was dripped onto a TEM grid and the excess liquid was removed after a short waiting time.
- The sample was dried on air and then transferred into the TEM.
- The investigation of the particles was done in TEM brightfield mode. Diffraction mode was used to check for crystallinity.

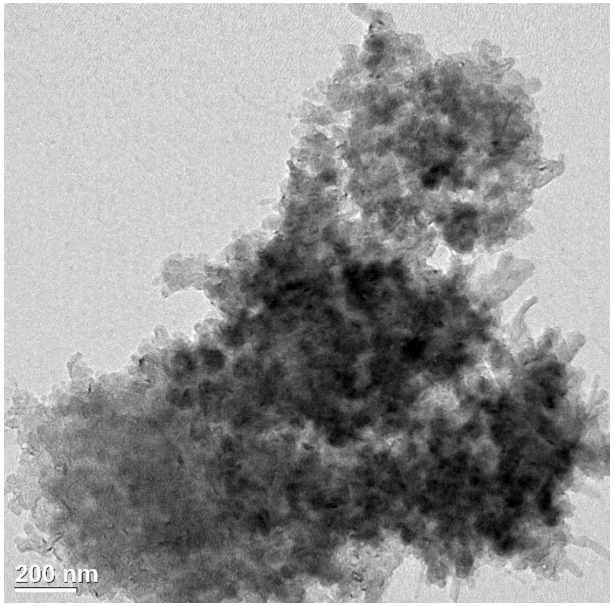
- **Note:** For high resolution imaging (> 100 kx), the samples on the grid were to large. To improve this, the grinding procedure and also the application of the sample to the grid should be improved.

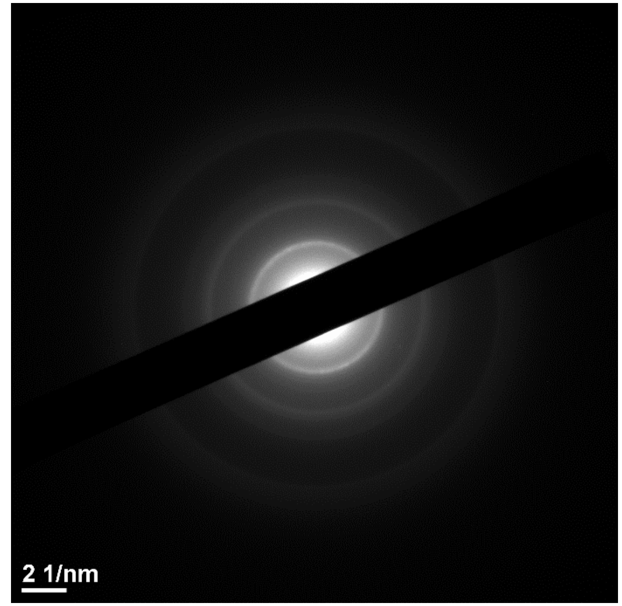
TEM investigation
Sample P01 (MB2955A2)



- Already in this low magnification (10.000 times), the porous structure of the particles can be seen.
- The particle is much larger than 1 µm in diameter. In the middle, it is hardly electron transparent, but at the edges, it can quite easily be investigated.

TEM investigation
Sample P01 (MB2955A2)





- The right SAED pattern was taken in the area shown in the left image.
- Diffraction rings can be seen, indicating that the particle is at least in parts polycrystalline. However, the rings are not that sharp. This might be because the particle also contains amorphous material.

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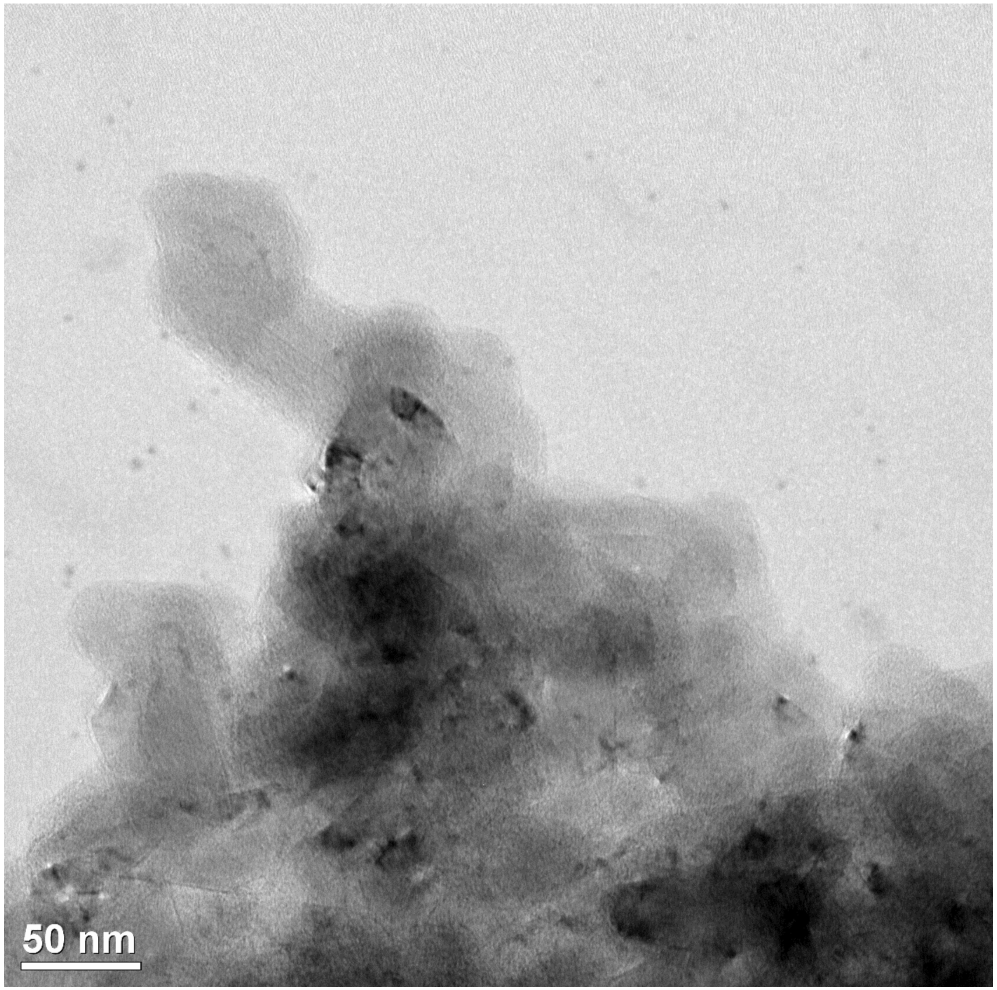
TEM investigation
Sample P01 (MB2955A2)



- This TEM image shows the and area at the edge of the particle in 50.000 times magnification.
- First hints of crystalline areas can already be seen in the image (dark regions).
- It turned out that the carbon support of the chosen TEM grid was too thick for high resolution imaging. This needs to be improved next time by choosing a different TEM grid type.

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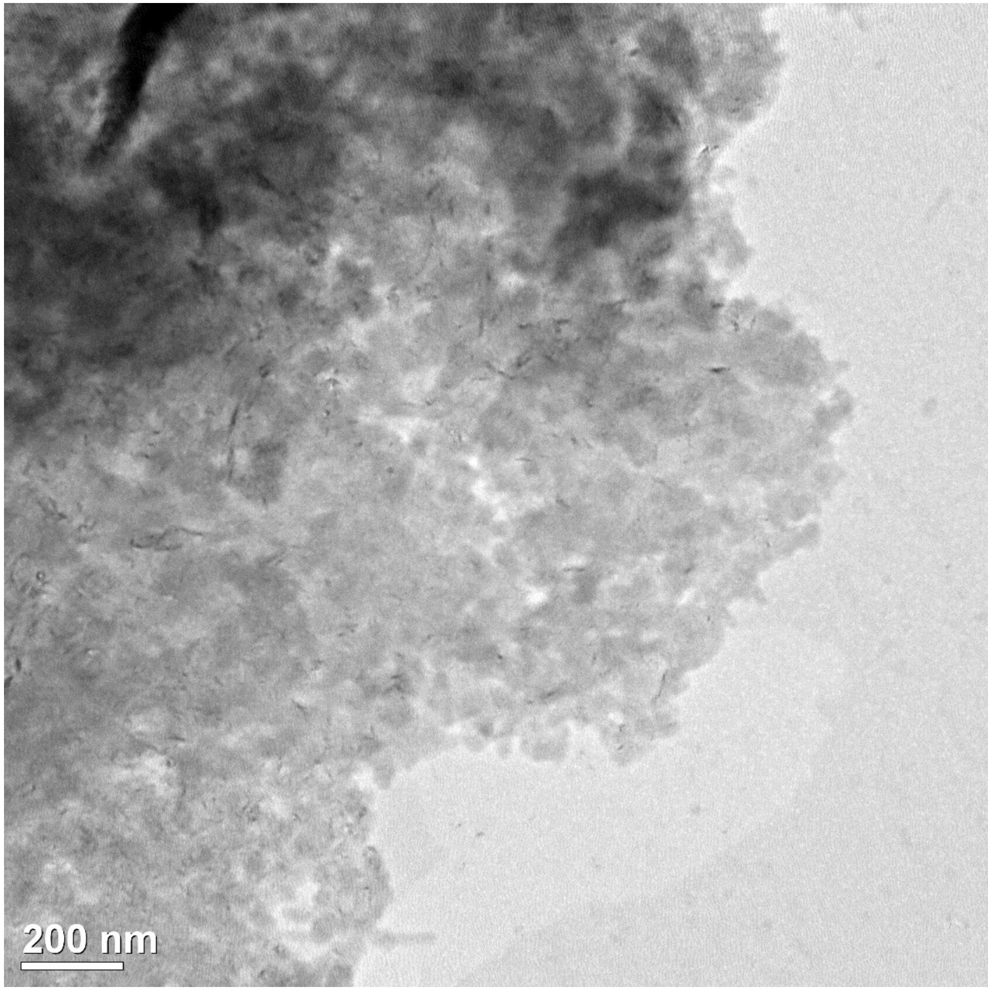
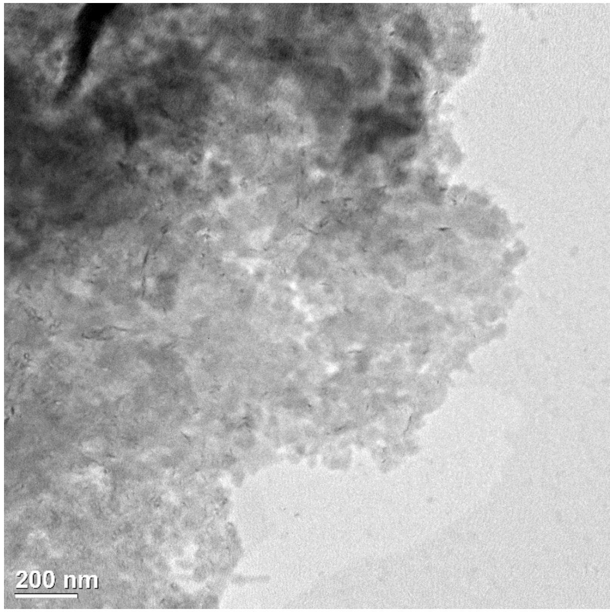
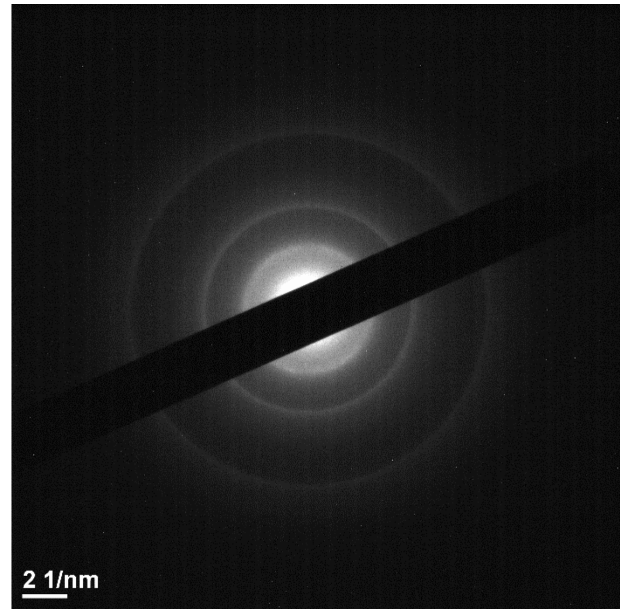
TEM investigation
Sample P01 (MB2955A2)



- This image in 10.000 times maginification shows another particle.
- Also here, the posrous structure can already be seen in this relatively low magnification.

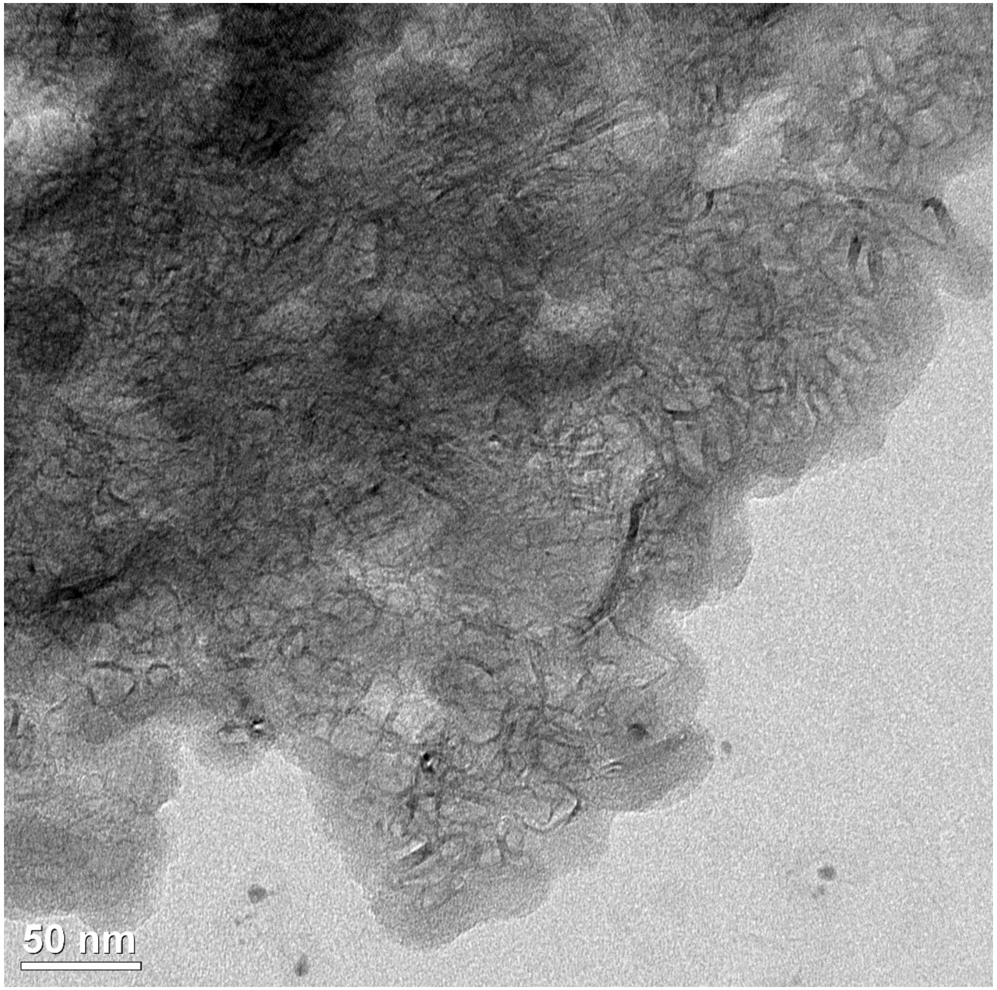
TEM investigation
Sample P01 (MB2955A2)





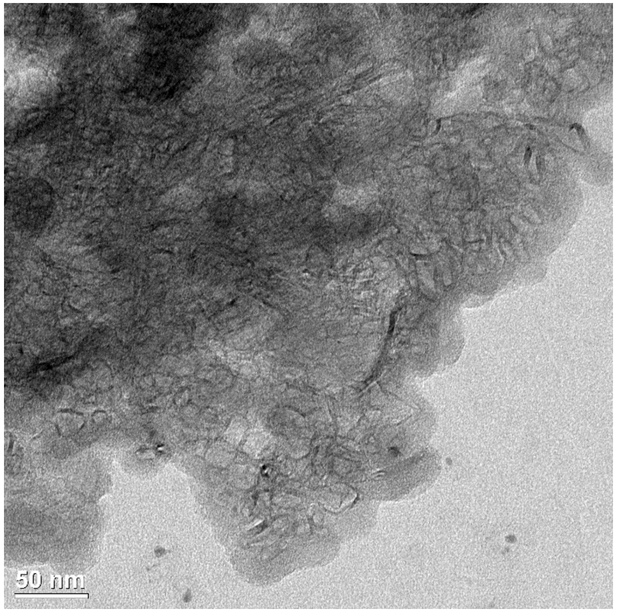
- The right SAED pattern was taken in the area shown in the left image.
- Diffraction rings can be seen, indicating that the particle is at least in parts polycrystalline. However, the rings are again not that sharp. This might be because the particle also contains amorphous material.

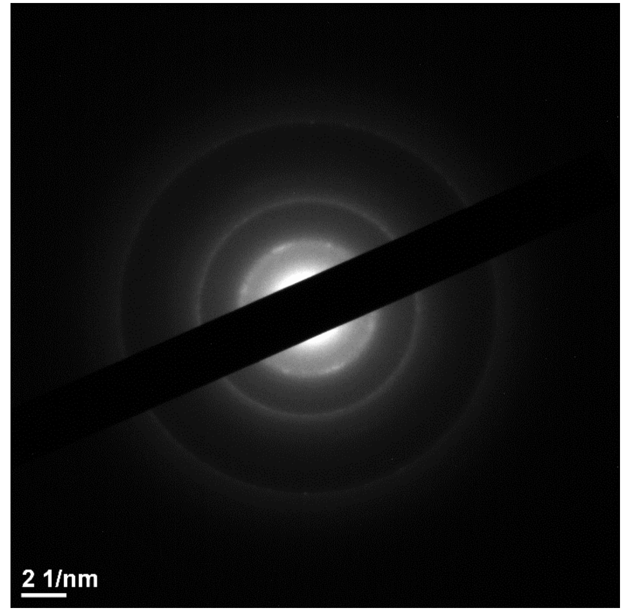
TEM investigation
Sample P01 (MB2955A2)



- This TEM image shows the and area at the edge of the particle in 50.000 times magnification.
- Pores in the material can clearly be seen.

TEM investigation
Sample P01 (MB2955A2)





- The right SAED pattern was taken in the area shown in the left image.
- Diffraction rings can be seen, indicating that the particle is at least in parts polycrystalline. Some diffration spots in the rings can be seen. These stem from small crystals in the area of investigation.

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Final Remarks

- The analyses are carried out by specifically qualified employees and by the use of up-to-date analysis machines. Nevertheless, we are not able to provide guarantee for the correctness of the drawn conclusion.
- The results only refer to the provided and examined samples.
- Sample taking and shipping of the samples to the ZAA took place under the responsibility of the client.
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End of report

Total number of pages: 13